OMKAR VAIDYA

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WORK EXPERIENCE

BrainChain Pvt. Ltd. **Sep 2025 – Jul 2026**
Software Engineer Intern (LLM & SDE) *Pune, India*

- Engineered an **AI-powered GRE vocabulary tutor** utilizing **NLP, LangChain**, and a fine-tuned **DistilGPT-2** model for personalized learning modules.
- Integrated a **LangChain orchestration layer** connecting **FastAPI** and **Spring Boot** services with a responsive **ReactJS** user interface.
- Built automated data ingestion and preprocessing pipelines within a **4-member Agile team**, participating in code reviews and bi-weekly production sprints.

FEV India Pvt. Ltd. **Jun 2025 – Sep 2025**
Data Science Intern (Project Intern) *Pune, India*

- Developed real-time **EV analytics systems** encompassing **State of Charge (SoC)**, range estimation, and **ETA forecasting** to optimize vehicle performance.
- Built an **AI-driven charging recommendation engine** by analyzing driving and charging telemetry to enhance battery longevity and route efficiency.
- Designed **trip post-analytics pipelines** and interactive **Python, Dash, and Plotly** dashboards for driver efficiency scoring and live telemetry monitoring.

PROJECTS

Auto Worksite Safety Monitoring | *Python, YOLOv8, Deep Learning, OpenCV, Tkinter* **Jul 2024 – Aug 2025**

- Developed a custom **YOLOv8 architecture** integrated with **CBAM** and **ASFF** attention modules for high-precision **PPE detection** on construction sites.
- Designed an **IoU-based spatial matching algorithm** to associate detected PPE bounding boxes with workers and dynamically classify compliance as **SAFE** or **UNSAFE**.
- Built an interactive **Tkinter GUI** for real-time video stream inference, automated PPE labeling, and live safety compliance reporting.

Staff Scheduler for Part-Time Jobs | *Python, Machine Learning, Scikit-learn, Streamlit* **Dec 2024 – Feb 2025**

- Engineered an **ML-driven scheduling optimization tool** analyzing footfall traffic, shift workloads, and employee availability to automate shift allocation.
- Implemented and tuned a multi-layer perceptron (**MLP Regressor**) on feature-engineered data to model and predict **employee satisfaction scores**.
- Deployed an interactive **Streamlit web dashboard** integrating serialized pipelines and scalers for real-time inference, shift generation, and roster visualization.

Anti-Sleep Driver Alert System | *C++, Arduino, Embedded IoT, IR Sensors* **Sep 2022 – Jan 2023**

- Engineered a **wearable IoT safety device** for real-time driver drowsiness detection and automated alerting using **infrared (IR) blink-tracking sensors**.
- Programmed an **Arduino microcontroller** with low-latency interrupt logic to trigger a multi-stage buzzer and alert mechanism, mitigating road accident risks.
- Tested and calibrated sensor thresholds across varied ambient lighting conditions to ensure reliable eye-closure tracking and minimal false positives.

TECHNICAL SKILLS

- **Languages:** Python, Java, C++, SQL, JavaScript
- **Frameworks & Libraries:** FastAPI, Spring Boot, ReactJS, LangChain, Scikit-learn, Streamlit, OpenCV, Dash, Plotly
- **Developer Tools & Databases:** Git, GitHub, MySQL, REST APIs, VS Code
- **Core Competencies:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), DBMS, Machine Learning, Operating Systems

EDUCATION

Pimpri Chinchwad College of Engineering (PCCOE), Pune **2022 – 2026**
Bachelor of Technology in Computer Engineering **Percentage: 77%**

ASM CSIT College of Commerce, Science and IT, Pune **2021**
Higher Secondary Certificate (HSC) **Percentage: 77%**

LEADERSHIP, ACTIVITIES & PROFILES

Technical Head, PCCOE ACM Student's Chapter *Sept 2024 – Aug 2025*

Web Developer, PCCOE ACM Student's Chapter *Sept 2023 – Aug 2024*